```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score

# Sample dataset
data = {
    'Hours_Studied': [1, 2, 2, 3, 4, 5, 5, 6, 7, 8],
    'Passed':       [0, 0, 0, 0, 1, 1, 1, 1, 1, 1]
}

df = pd.DataFrame(data)

print(df)
```

```
   Hours_Studied  Passed
0              1       0
1              2       0
2              2       0
3              3       0
4              4       1
5              5       1
6              5       1
7              6       1
8              7       1
9              8       1
```

```python
X = df[['Hours_Studied']]
y = df['Passed']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = LogisticRegression()

model.fit(X_train, y_train)
```

```
LogisticRegression()
```

```python
y_pred = model.predict(X_test)

print("Predicted values:", y_pred)
print("Actual values:   ", y_test.values)
```

```
Predicted values: [1 0 1]
Actual values:    [1 0 1]
```

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 1.0
```

```
cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:")
print(cm)
```

```
Confusion Matrix:
[[1 0]
 [0 2]]
```

```
sns.heatmap(cm, annot=True, cmap='Blues', fmt='d')

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.show()
```

